The primal problem is

$$\min_{x \in \mathbb{R}^2} -4x_1 + 2x_2$$

subject to

$$\begin{aligned} -x_1 + x_2 &= 2, \\ x_1 - x_2 &= 1, \\ x_1 &\geq 0, \\ x_2 &\geq 0. \end{aligned}$$

The two equality constraints are incompatible, and the optimization problem is infeasible. Consequently, we expect the dual problem to be either infeasible or unbounded.

To write the Lagrangian, we first write the problem as follows:

$$\min_{x \in \mathbb{R}^2} -4x_1 + 2x_2$$

subject to

$$\begin{aligned} h_1(x) &= 2 + x_1 - x_2 = 0 \quad (\lambda_1), \\ h_2(x) &= 1 - x_1 + x_2 = 0 \quad (\lambda_2), \\ g_1(x) &= \quad - x_1 \qquad \leq 0 \quad (\mu_1), \\ g_2(x) &= \qquad\qquad - x_2 \leq 0 \quad (\mu_2), \end{aligned}$$

where λ_1, λ_2, μ_1 and μ_2 are the penalty parameters associated with each constraint.

The Lagrangian function is

$$\begin{aligned} L(x_1, x_2, \lambda_1, \lambda_2, \mu_1, \mu_2) &= -4x_1 + 2x_2 + \lambda_1(2 + x_1 - x_2) \\ &\quad + \lambda_2(1 - x_1 + x_2) - \mu_1 x_1 - \mu_2 x_2 \\ &= (-4 + \lambda_1 - \lambda_2 - \mu_1)x_1 \\ &\quad + (2 - \lambda_1 + \lambda_2 - \mu_2)x_2 + 2\lambda_1 + \lambda_2. \end{aligned}$$

The dual function is

$$q(\lambda_1, \lambda_2, \mu_1, \mu_2) = \min_{x_1, x_2} L(x_1, x_2, \lambda_1, \lambda_2, \mu_1, \mu_2).$$

In order for the dual function to be bounded, the coefficients of x_1 and x_2 have to be zero. Therefore, we have to impose the following constraints on the penalty parameters:

$$\mu_1 = -4 + \lambda_1 - \lambda_2, \tag{1}$$

$$\mu_2 = 2 - \lambda_1 + \lambda_2. \tag{2}$$

In that case, the dual function becomes

$$q(\lambda_1, \lambda_2, \mu_1, \mu_2) = 2\lambda_1 + \lambda_2.$$

Moreover, the parameters associated with the inequality constraints must be non negative. That is, $\mu_1 \geq 0$ and $\mu_2 \geq 0$. The dual problem is therefore

$$\max_{\lambda_1, \lambda_2, \mu_1, \mu_2} 2\lambda_1 + \lambda_2$$

subject to

$$\begin{aligned}
\mu_1 &= -4 + \lambda_1 - \lambda_2, \\
\mu_2 &= 2 - \lambda_1 + \lambda_2, \\
\mu_1 &\geq 0, \\
\mu_2 &\geq 0.
\end{aligned}$$

As the parameters μ_1 and μ_2 do not appear in the objective function, we can rewrite (1) and (2) as

$$\begin{aligned}
-4 + \lambda_1 - \lambda_2 &\geq 0, \\
2 - \lambda_1 + \lambda_2 &\geq 0,
\end{aligned}$$

and eliminate the μ's, so that the dual problem becomes

$$\max_{\lambda_1,\lambda_2} 2\lambda_1 + \lambda_2$$

subject to

$$\begin{aligned}
\lambda_1 - \lambda_2 &\geq 4, \\
\lambda_1 - \lambda_2 &\leq 2.
\end{aligned}$$

The quantity $\lambda_1 - \lambda_2$ cannot be both larger or equal to 4, and lesser or equal to 2. Therefore, the dual problem is also infeasible.